DATASET DESCRIPTION

Fetal Health Classification — Dataset Reference

CTG Recordings • 21 Numeric Features • 2,126 Records • UCI ML Repository

Dataset File	fetal_health.csv
Location in Project	dataset/fetal_health.csv
Source	UCI Machine Learning Repository
Total Records	2,126 CTG study cases
Predictor Features	21 numeric CTG features
Target Column	fetal_health (1 = Normal, 2 = Suspect, 3 = Pathological)
Missing Values	None found during initial inspection
Feature Types	All numeric - no encoding needed

1. What This Dataset Is

The dataset contains measurements taken from cardiotocogram (CTG) recordings. A CTG is a test done during pregnancy to monitor the fetal heart rate and uterine contractions at the same time. Each row in the dataset represents one CTG study case, and the columns describe different aspects of what was recorded - things like how fast the fetal heart was beating, how many times it sped up or slowed down, and various statistical features of the heart rate pattern over time.

The data comes from the UCI Machine Learning Repository and was originally labelled by expert obstetricians who classified each recording as Normal, Suspect, or Pathological. That label is what the model is trained to predict.

The dataset is clean and entirely numeric, which makes it straightforward to work with. There is no messy text to parse or categories to encode - the values can be loaded and used almost directly.

2. Target Label and Class Distribution

The target column is called fetal_health and contains one of three values:

Value	Class	Count	Share of Dataset
1.0	**Normal**	1,655	~78%
2.0	**Suspect**	295	~14%

3.0	**Pathological**	176	~8%

The imbalance here is worth keeping in mind. The model will see far more Normal examples during training than Suspect or Pathological ones. This can cause it to be better at predicting Normal and worse at catching the rarer classes - which is actually the opposite of what matters most clinically. This is something to watch when looking at the evaluation results.

3. The 21 Input Features

The features are split into four groups based on what they measure. All of them come directly from the CTG recording.

3.1 Baseline and Movement Features

These describe the general state of the fetal heart rate and any movement activity recorded during the session.

Feature Name	What it means
`baseline value`	The average baseline fetal heart rate in beats per minute
`accelerations`	How many times the heart rate sped up briefly during the recording
`fetal_movement`	Amount of fetal movement detected during the session
`uterine_contractions`	How much contraction activity occurred during the recording

3.2 Deceleration Features

These track how often the fetal heart rate dropped during the recording and how serious those drops were.

Feature Name	What it means
`light_decelerations`	Number of mild, brief drops in fetal heart rate
`severe_decelerations`	Number of more serious heart rate drops that may indicate distress
`prolongued_decelerations`	Number of decelerations that lasted for an extended period

3.3 Variability Features

Variability refers to how much the fetal heart rate fluctuates over time. Both too little and too much variability can be a sign that something is wrong.

Feature Name	What it means
abnormal_short_term_variability	Percentage of time showing abnormal short-term heart rate variation
mean_value_of_short_term_variability	Average amount of short-term variation in the heart rate
percentage_of_time_with_abnormal_long_term_variability	How much of the recording had abnormal long-term variability
mean_value_of_long_term_variability	Average amount of long-term variation in the heart rate

3.4 Histogram Features

These come from building a histogram of the fetal heart rate values over the recording. They describe the shape and spread of that distribution.

Feature Name	What it means
histogram_width	The range between the lowest and highest values in the histogram
histogram_min	The minimum heart rate value in the histogram
histogram_max	The maximum heart rate value in the histogram
histogram_number_of_peaks	How many peaks (high points) appear in the histogram shape
histogram_number_of_zeroes	Number of bins in the histogram with zero count
histogram_mode	The most frequently occurring heart rate value
histogram_mean	The average heart rate across the whole distribution
histogram_median	The middle value of the heart rate distribution
histogram_variance	How spread out the heart rate values are
histogram_tendency	Whether the distribution leans left, right, or is roughly symmetric

4. Data Quality and Preparation

The dataset is in good shape and did not need a lot of cleaning. Here is a summary of what was checked and what was done:

- No missing values were found during the initial inspection. All 2,126 rows have values for all 21 features.
- All feature columns are numeric, so there is no need to encode any categories or convert text to numbers.
- The feature values are already in their natural ranges from the original CTG recordings. No scaling was applied since Random Forest does not require it.
- The column order in the dataset is the same order that must be used in the web form. If the inputs come in the wrong order, the predictions will be wrong even if the values themselves are correct.

The only thing to be aware of is the class imbalance mentioned earlier. The dataset was not resampled or artificially balanced for this version of the project, so the model training reflects the natural distribution of the data.

5. How the Dataset is Used in the Project

The dataset plays two roles in this project:

- Training and evaluation: the CSV is loaded in the Jupyter notebook, split into training and test sets, and used to train and evaluate the Random Forest model.
- Defining the input schema: the 21 feature names and their order in the CSV are used directly to build the web form. When a user fills in the form and submits it, the values are mapped into a DataFrame using the same column names as the training data before being passed to the model.

This means the dataset file is only needed during training. Once the model is saved as a .pkl file, the app does not need to load the CSV anymore - it just needs the trained model.

```
dataset/fetal_health.csv  →  used during training only
optimized_fetal_health_model.pkl  →  used by the app at runtime
```

6. Limitations and Intended Use

A few things are worth being clear about when it comes to this dataset:

- It is a research and teaching dataset, not a certified clinical resource. The labels were assigned by experts but the data has not been validated for use in real medical decision-making.
- The class imbalance means that accuracy alone is not a good measure of how well the model works. A model that always predicts Normal would still get about 78% accuracy just from that - which is why recall for Suspect and Pathological cases is the more important number to look at.
- The dataset covers a specific set of CTG features from a particular study. It may not represent all possible CTG patterns or patient populations.

- For this project, the dataset is used to demonstrate how machine learning can be applied to healthcare-related data. Any predictions made by the app should be treated as a learning exercise, not medical advice.

End of Dataset Description Document